Science practice booklet: Grade 6 expectations

Grade 6 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 6 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Living things

Standards practised: MS-LS1-1, MS-LS1-2, MS-LS1-3, MS-LS1-6, MS-LS2-1, MS-LS2-3

A worked example

A mangrove leaf is made of cells. Name the part of a leaf cell that captures light for photosynthesis, and say what the leaf makes and gives off.

1. Plant cells have green parts called chloroplasts. They contain a green substance that captures light energy.
2. In photosynthesis the plant uses light energy, water and carbon dioxide.
3. These are changed into sugar (food for the plant) and oxygen.
4. The sugar is used or stored by the plant, and the oxygen is released into the air.

Chloroplasts capture the light; the leaf makes sugar and gives off oxygen.

Questions

1. What is the basic unit of structure and function in all living things?

2. What is the job of the cell membrane?

3. Which part of a cell holds the instructions that control the cell's activities?

4. Name two parts that a plant cell has but an animal cell does not.

5. Mitochondria are called the 'powerhouses' of the cell. What do they do?

6. Write the inputs and outputs of photosynthesis in words.

7. Put these in order from smallest to largest: organ, cell, organ system, tissue.

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8. Match each body system to its job: circulatory, respiratory, digestive. (Jobs: breaks down food; brings oxygen in; carries blood around the body.)

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9. Hamdan runs in the school yard and his breathing and heartbeat both get faster. Explain how two body systems work together here.

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10. In a mangrove food web, plants make food, crabs eat leaves and herons eat crabs. Which organisms have the most energy available, and why does the amount get smaller along the chain?

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11. Oryx and gazelles in a desert reserve share a small number of water holes and plants. What can happen to their numbers if a very dry year reduces the plants?

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12. A seedling is kept in a dark cupboard for two weeks and turns pale and droopy. Use photosynthesis to explain why.

Section 2: Earth and space

Standards practised: MS-ESS2-1, MS-ESS2-2, MS-ESS2-3, MS-ESS2-4, MS-ESS2-6

A worked example

Seawater in the Gulf warms in the sun, and later rain falls on the Hajar mountains. Describe the steps of the water cycle that connect these two events and say what powers the cycle.

1. The Sun warms the sea and some of the water evaporates, turning into water vapour (a gas).
2. The warm air rises and cools. The vapour condenses into tiny drops that form clouds.
3. When the drops get big enough they fall as precipitation, such as rain.
4. The Sun's energy and gravity keep the water moving around the cycle.

Evaporation, condensation, then precipitation; the Sun's energy and gravity power the cycle.

Questions

1. Name the three main steps of the water cycle.

2. Water drops form on the outside of a cold glass of juice on a humid day. Which step of the water cycle is this like, and why?

3. Which two things move water around the water cycle?

4. What is the difference between weather and climate?

5. Is 'It rained in Fujairah this morning' a statement about weather or climate? Is 'Summers in the UAE are hot and dry' about weather or climate?

6. List Earth's four main layers from the outside to the centre.

7. Which layer of the Earth do we live on, and is it the thinnest or the thickest layer?

8. Earth's crust is broken into large pieces that move a few centimetres each year. What are these pieces called, and name one event they can cause where they meet.

9. Scientists find sea shell fossils high in the Hajar mountains. What does this suggest about that place long ago?

10. Name the three main types of rock.

11. Wind blows sand and builds a dune, and over millions of years layers of sand are pressed and cemented into rock. Which type of rock forms, and which process moved the sand?

12. Explain why the rock cycle never ends. Give an example of one rock changing into another type.

Section 3: Matter, forces and energy

Standards practised: MS-PS1-4, MS-PS3-3, MS-PS3-4, MS-PS2-2

A worked example

Noura puts an ice cube in a cup in a warm kitchen. Describe what happens to the particles of water as the ice melts and then the water warms up.

1. In ice (a solid) the particles are packed closely and only vibrate in place.
2. Heat flows from the warm room into the ice, and the particles gain energy and move more.
3. They loosen and slide past each other: the ice melts into liquid water.
4. As the liquid keeps getting heat, the particles move faster and the temperature rises.

The particles gain energy, move more and slide past each other (melting), then move faster as the water warms.

Questions

1. Describe how the particles are arranged and move in a solid, a liquid and a gas.

2. What happens to the particles of a substance when it is heated?

3. At what temperature does pure water boil and at what temperature does it freeze, in degrees Celsius?

4. When a puddle dries up in the sun, which change of state has happened?

5. A football left in a hot car in the afternoon sun feels harder and tighter. Use particles to explain why.

6. Khalid holds a hot cup of karak. In which direction does heat flow between the cup and his hands?

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7. The sand on a UAE beach is too hot to walk on at noon. How does the Sun's energy reach the sand, and which way of heat transfer is it: conduction, convection or radiation?

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8. Which two pairs are matched correctly? (a) A metal spoon in hot soup warms up: conduction. (b) Warm air rising above a heater: convection. (c) A heater warming the floor through the air: conduction.

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9. Hind wants her cold drink to stay cold on a hot day. Choose a material for the container and explain your choice.

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10. A bathtub of water and a cup of water are both at 30 °C. Which has more thermal energy, and why?

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11. Solar panels in a UAE solar park change the Sun's light energy into another form of energy. Which form?

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12. Two teams pull on a rope with equal and opposite forces and the rope does not move. Are the forces balanced or unbalanced? What would happen if one team pulled harder?

Answer guide for parents

Section 1: Living things

1. The cell.
2. It surrounds the cell and controls what goes in and out, such as water, food and waste.
3. The nucleus.
4. A cell wall and chloroplasts (plant cells also have a large vacuole).
5. They release energy from food so the cell can use it.
6. Carbon dioxide + water, using light energy, make sugar (glucose) + oxygen.
7. Cell, tissue, organ, organ system.
8. Circulatory carries blood around the body; respiratory brings oxygen in; digestive breaks down food.
9. His muscles need more oxygen. The respiratory system takes in more oxygen and the circulatory system (heart and blood) carries it to the muscles faster.
10. The mangrove plants (producers). At each step much of the energy is used by the organism for living and is lost as heat, so less is passed on.
11. There is less food, so the animals compete more and the populations may get smaller. Ecosystems depend on resources being available.
12. Without light the plant cannot do photosynthesis, so it cannot make sugar for energy and growth, and its leaves lose their green colour.

Section 2: Earth and space

1. Evaporation, condensation and precipitation.
2. Condensation: water vapour in the air touches the cold glass, cools and turns into liquid water.
3. Energy from the Sun and gravity.
4. Weather is the conditions in one place over a short time (today or this week). Climate is the pattern of weather over many years.
5. The first is weather. The second is climate.
6. Crust, mantle, outer core, inner core.
7. The crust, which is the thinnest layer.
8. Tectonic plates. Where they meet they can cause earthquakes, volcanoes or mountains. (Any one.)
9. It was once under the sea, and over a very long time the land moved and was lifted up. Earth's surface changes.
10. Igneous, sedimentary and metamorphic.
11. Sedimentary rock forms. The wind moved the sand (erosion, then deposition where it settled).
12. Rocks are always being worn away, melted, pressed or heated and changed. For example, an igneous rock can be broken into sand, which is pressed into sedimentary rock.

Section 3: Matter, forces and energy

1. Solid: packed closely, vibrating in place. Liquid: close together but sliding past each other. Gas: far apart, moving freely in all directions.
2. They gain energy and move faster (and the temperature rises).
3. It boils at 100 °C and freezes at 0 °C.
4. Evaporation: liquid water changes to water vapour (a gas).
5. The air particles inside gain energy and move faster, so they hit the inside of the ball harder and more often and push outwards more.
6. From the hot cup to his cooler hands (heat always flows from hotter to colder).
7. It travels through space and air as light and heat rays. This is radiation.
8. (a) and (b). In (c) the heat reaches the floor through the air, so it is not conduction.
9. A good insulator, such as a foam cooler or a vacuum flask. It reduces the heat flowing in from the hot air outside.
10. The bathtub. It has more water, so more particles carrying energy, even though the temperature is the same.
11. Electrical energy (electricity).
12. Balanced. If one team pulled harder the forces would be unbalanced and the rope would start moving towards that team.